INCH - POUND

MS20028F
20 May 2005
SUPERSEDING
MS20028E (USAF)
16 August 1976

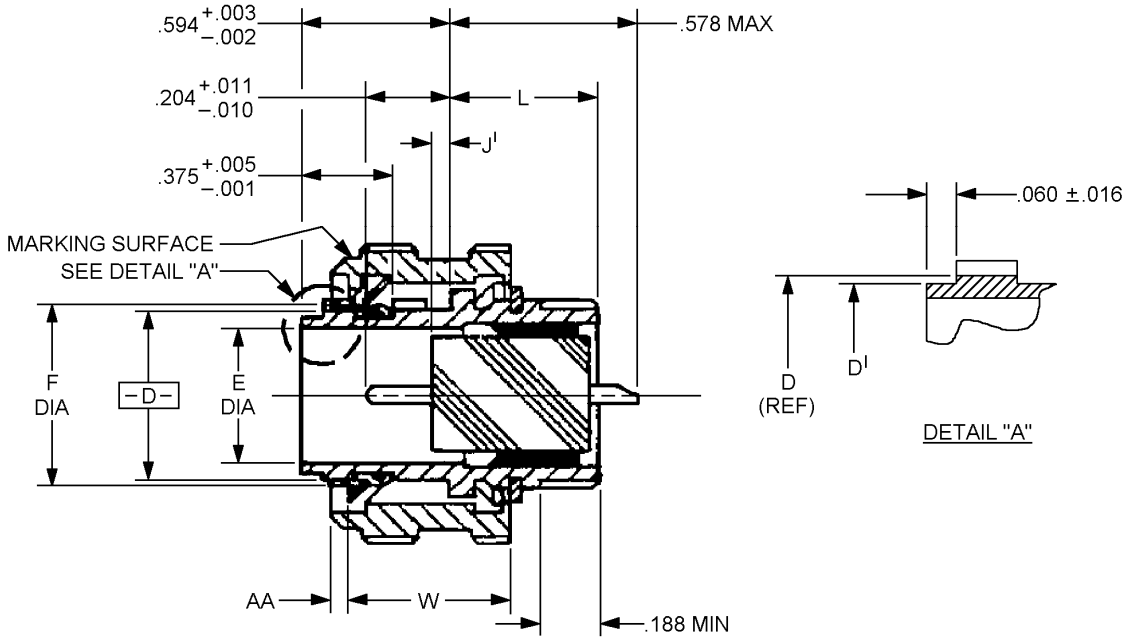
DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, STRAIGHT, SOLDER TYPE, BAYONET COUPLING, CLASS T, SERIES I

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 7 December 1998.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27599.



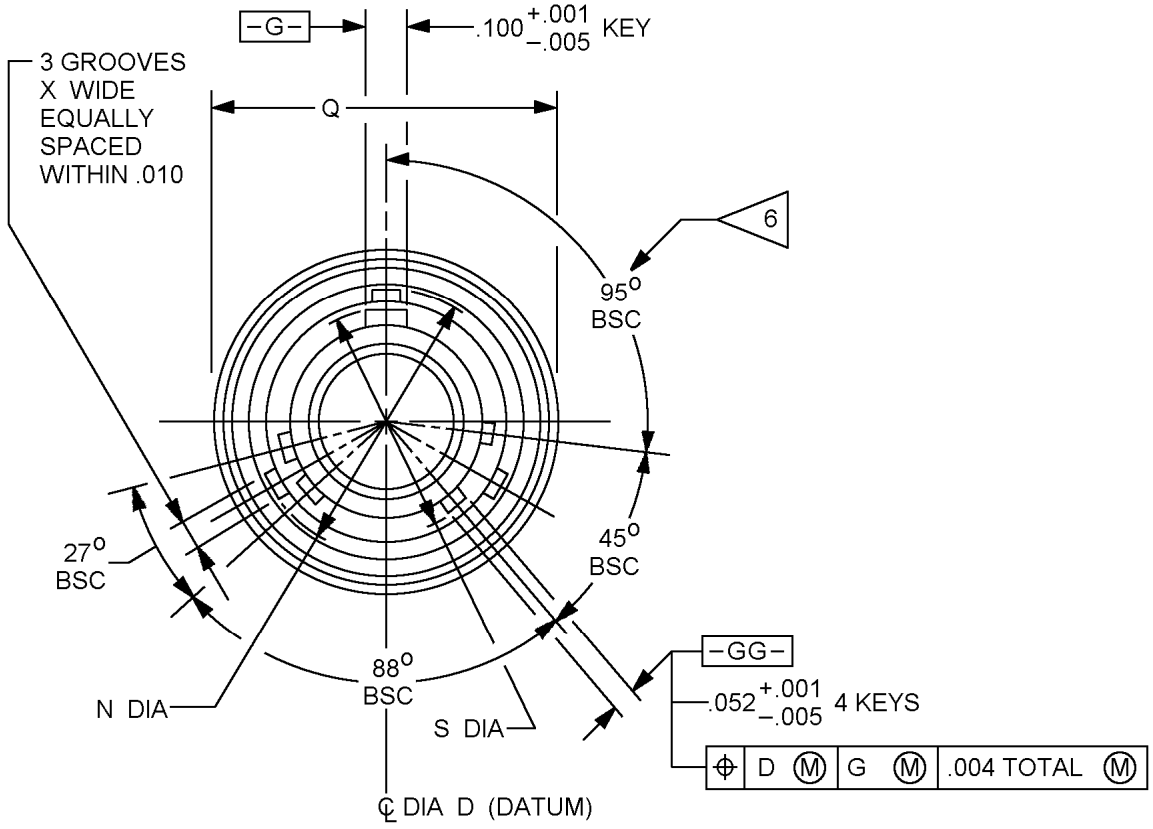
CLASS T

FIGURE 1. Plug.

AMSC N/A

FSC 5935

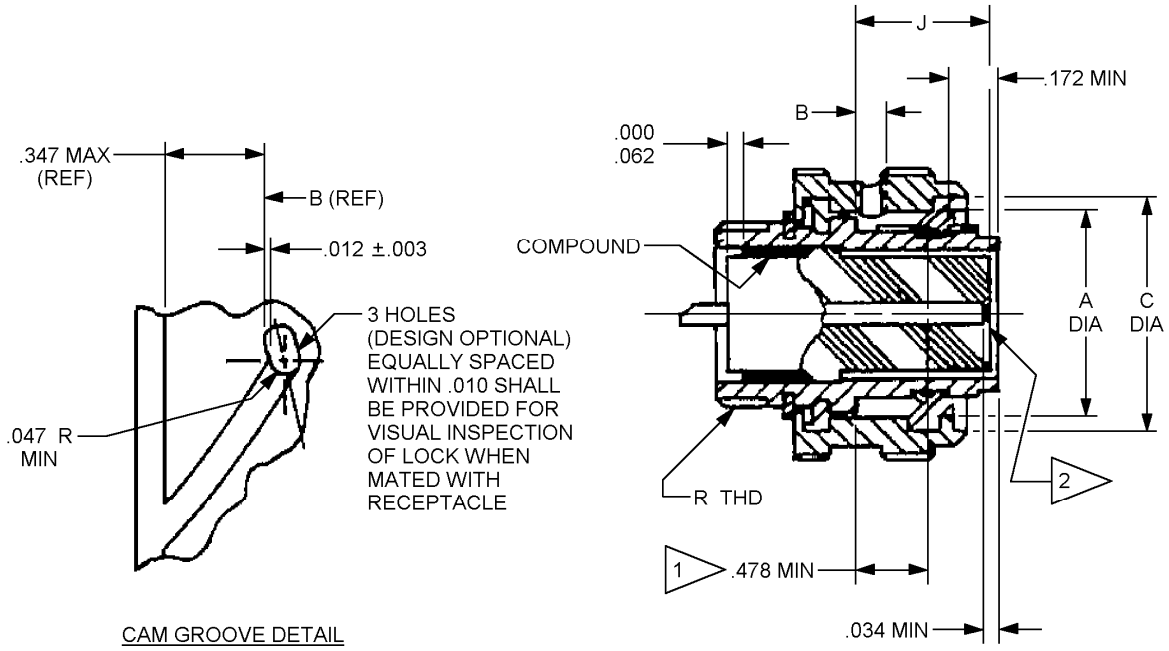
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.



CLASS T

FIGURE 1. Plug - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.



CLASS T

NOTES:

1. The point at which a gauge pin having the same basic diameter as the mating contact and a square face, touches socket contact spring.
2. Insert front surface shall be flat within .005 T.I.R.
3. Diameter D with respect to diameter S shall be concentric at MMC.
4. Diameters A and N shall be concentric within .008 T.I.R.
5. Dimensions are in inches. Metric equivalents are in parenthesis and given for information only.
6. Normal keyway position. For other keyway positions, see MIL-DTL-27599.
7. Finish: A - Bright cadmium (see MIL-DTL-27599).
   B - Olive drab (see MIL-DTL-27599).
   F - Electroless nickel in accordance with SAE-AMS-C-26074, class 3 or 4, grade B, finish dull.

FIGURE 1. Plug - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.0254	0.011	0.280	0.06	1.524	0.347	8.814
0.002	0.051	0.012	0.305	0.062	1.575	0.375	9.525
0.003	0.076	0.016	0.406	0.100	2.54	0.478	12.14
0.004	0.102	0.034	0.864	0.172	4.370	0.578	14.68
0.005	0.127	0.047	1.194	0.188	4.775	0.594	15.09
0.010	0.254	0.052	1.321	0.204	5.182		

FIGURE 1. Plug - Continued.

TABLE 1. Dimensions.

Shell size	A dia + .005 - .001	B ± .020	C dia + .005 - .001	D dia + .001 - .005	D[1] dia min	E dia ± .003	F dia min
9	.560 (14.22)	.113 (2.87)	.684 (17.37)	.437 (11.10)	.417 (10.60)	.315 (8.00)	.447 (11.35)
11	.712 (18.08)		.832 (21.13)	.565 (14.35)	.545 (13.84)	.433 (11.00)	.575 (14.60)
13	.861 (21.87)		.981 (24.92)	.677 (17.20)	.657 (16.68)	.547 (13.90)	.692 (17.57)
15	.985 (25.02)		1.106 (28.09)	.802 (20.37)	.782 (19.86)	.672 (17.07)	.817 (20.75)
17	1.110 (28.20)		1.230 (31.24)	.927 (23.54)	.907 (23.04)	.797 (20.24)	.942 (23.92)
19	1.217 (30.91)		1.347 (34.21)	1.032 (26.21)	1.012 (25.70)	.886 (22.50)	1.047 (26.60)
21	1.342 (34.08)		1.462 (37.13)	1.157 (29.38)	1.137 (28.88)	1.011 (25.68)	1.172 (29.76)
23	1.467 (37.26)		1.587 (40.31)	1.282 (32.56)	1.262 (32.05)	1.136 (28.85)	1.297 (32.94)
25	1.592 (40.44)	.152 (3.86)	1.712 (43.48)	1.407 (35.73)	1.387 (35.22)	1.260 (32.00)	1.442 (36.62)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.

TABLE 1. Dimensions - Continued.

Shell size	J ± .005 socket insert	J[1] ± .005 pin insert	L max	N dia + .005 - .006	Q dia max	S dia + .001 - .010
9	.576 (14.63)	.018 (.457)	.547 (13.90)	.674 (17.12)	.859 (21.82)	.482 (12.24)
11				.822 (20.87)	.984 (25.00)	.610 (15.49)
13				.972 (24.68)	1.156 (29.36)	.759 (19.27)
15				1.097 (27.86)	1.281 (32.53)	.884 (22.45)
17				1.222 (31.03)	1.406 (35.71)	1.009 (25.63)
19				1.329 (33.75)	1.516 (38.50)	1.114 (28.30)
21				1.454 (36.93)	1.641 (41.68)	1.239 (31.47)
23				1.579 (40.10)	1.766 (44.85)	1.364 (34.64)
25			.609 (15.47)	1.704 (43.28)	1.891 (48.03)	1.489 (37.82)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.

TABLE 1. Dimensions - Continued.

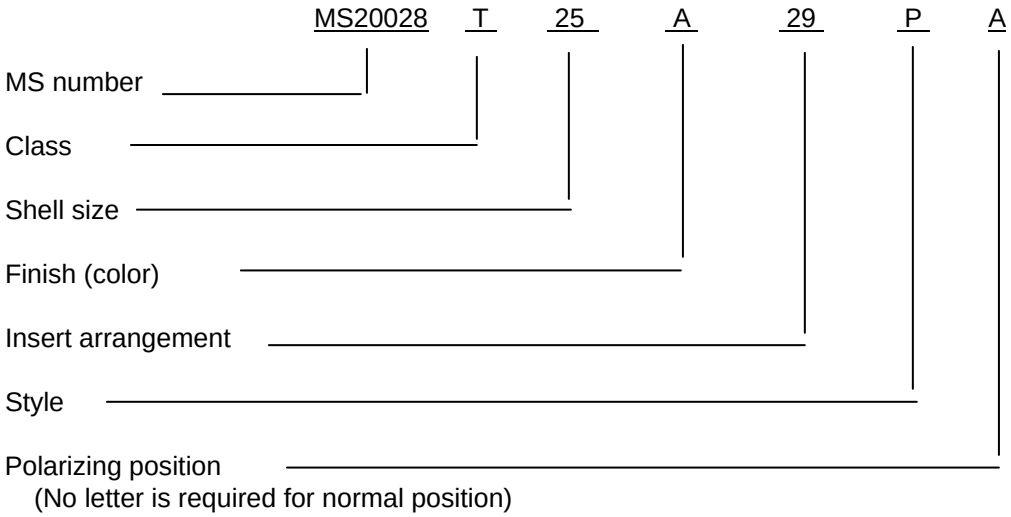
Shell size	W +.016 -.006	X +.016 -.001	R thread class 2A plated UNEF	AA +.005 -.006
9	.693 (17.60)	.090 (2.29)	7/16 - 28	.068 (1.73)
11	.682 (17.32)		9/16 - 24	
13			11/16 - 24	
15	.693 (17.60)		13/16 - 20	
17			15/16 - 20	
19			1 1/16 - 18	
21			1 3/16 - 18	
23			1 5/16 - 18	
25	.794 (20.16)	.137 (3.47)	1 7/16 - 18	.047 (1.20)

REQUIREMENTS

Dimensions and configurations: See figure 1 and table 1.
This connector mates with MS20026 and MS20029.
Insert arrangement shall be in accordance with MIL-STD-1560.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Reference documents. In addition to MIL-DTL-27599, this document references the following:

MIL-STD-1560
MS20026
MS20029
SAE-AMS-C-26074

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Air Force - 99
DLA – IS

(Project 5935-4648-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:21Z
Check the source to verify that this is the current version before use.